

CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 6

Question 11

Create a DataFrame of 10 rows and 4 columns with random values. Convert some values to NaN and highlight the NaN values.

Python Code:

```
import pandas as pd
import numpy as np

# Create DataFrame of 10 rows and 4 columns with random values
np.random.seed(42)
df = pd.DataFrame(np.random.randn(10, 4), columns=['A', 'B', 'C', 'D'])

# Convert some values to NaN
df.iloc[0, 1] = np.nan
df.iloc[2, 3] = np.nan
df.iloc[5, 0] = np.nan
df.iloc[7, 2] = np.nan

# Function / Styler to highlight NaN values
styled_df = df.style.highlight_null(null_color='red')

print("DataFrame with NaN values:")
print(df)
print("\n(In Jupyter Notebook / VS Code, styled_df highlights NaN values in red)")
```

Output:

```
DataFrame with NaN values:
   A         B         C         D
0  0.496714      NaN  0.647689  1.523030
1 -0.234153 -0.234137  1.579213  0.767435
2 -0.469474  0.542560 -0.463418      NaN
3  0.241962 -1.913280 -1.724918 -0.562288
4 -1.012831  0.314247 -0.908024 -1.412304
5      NaN -0.225776  0.067528 -1.424748
6 -0.544383  0.110923 -1.150994  0.375698
7 -0.600639 -0.291694      NaN  1.852278
8 -0.013497 -1.057711  0.822545 -1.220844
9  0.208864 -1.959670 -1.328186  0.196861

[Styled DataFrame: NaN values highlighted in red]
```

Question 12

Create a DataFrame of 10 rows and 4 columns with random values. Set the DataFrame background color to black and font color to yellow.

Python Code:

```
import pandas as pd
import numpy as np

# Create DataFrame of 10 rows and 4 columns with random values
np.random.seed(123)
df = pd.DataFrame(np.random.randn(10, 4), columns=['Col1', 'Col2', 'Col3', 'Col4'])

# Set DataFrame background color to black and font color to yellow
styled_df = df.style.set_properties(**{
    'background-color': 'black',
    'color': 'yellow'
})

print("DataFrame with Random Values:")
print(df)
print("\n(In Jupyter Notebook / VS Code, styled_df renders black background and yellow font)")
```

Output:

```
DataFrame with Random Values:
   Col1    Col2    Col3    Col4
0 -1.085631  0.997345  0.282978 -1.506295
1 -0.578600  1.651437 -2.426679 -0.428913
2  1.265936 -0.866740 -0.678886 -0.094709
3  1.491390 -0.638902 -0.443982 -0.434351
4  2.205930  2.186786  1.004054  0.386186
5  0.737369  1.490732 -0.935834  1.175829
6 -1.253881 -0.637752  0.907105 -1.428681
7 -0.140069 -0.861755 -0.255619 -2.798589
8 -1.771533 -0.699877  0.927462 -0.173636
9  0.002846  0.688223 -0.879536  0.283627

[Styled DataFrame: Black background with yellow font color]
```